

K-Field Galaxy-Plane Intersection Distribution Analysis

Randomized isotropic great-circle intersections on the unit sphere and the probability of a
20-plane concentration inside a 0.5-degree-diameter window

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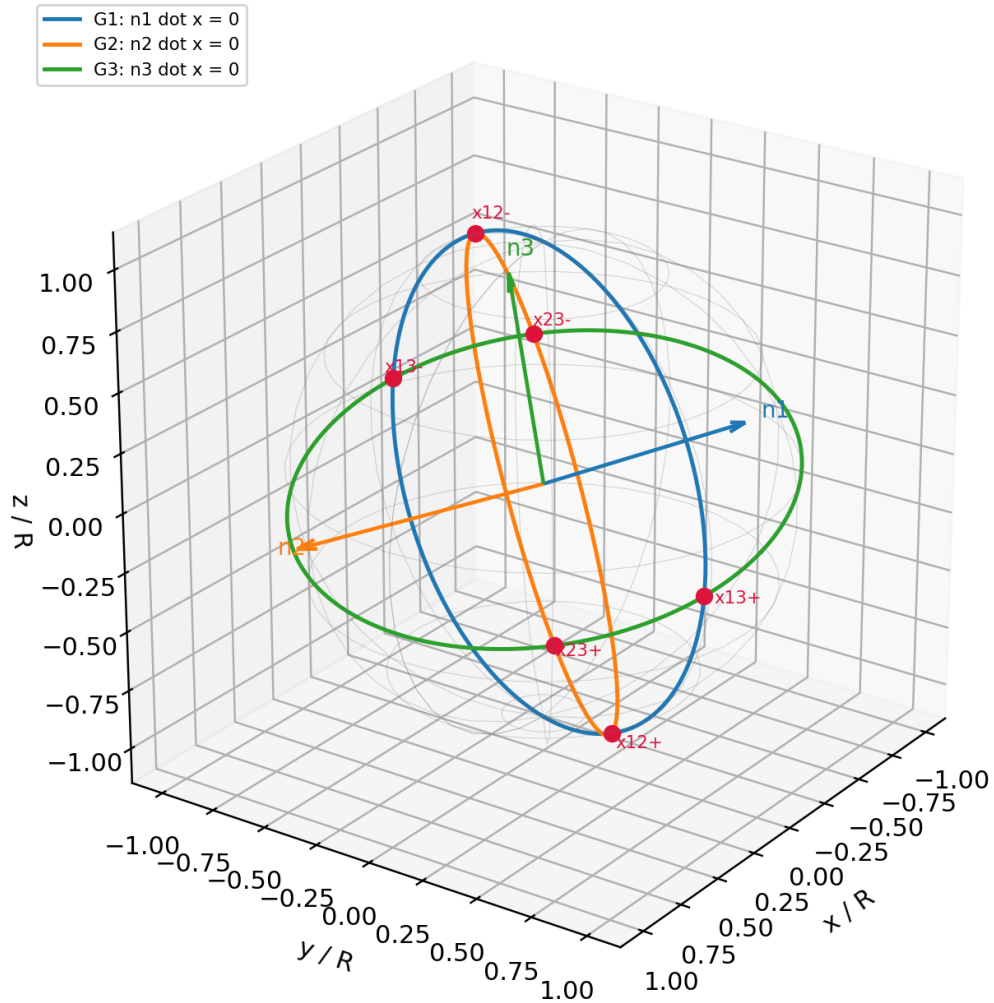
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Generated: Unix epoch 1784734315

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Base incidence cell: three randomized origin-centered planes on the unit sphere



Title figure. Dimensionally correct base incidence cell on a unit sphere of radius $R = 1$. The three displayed planes are the first three isotropic planes from the deterministic randomized realization with seed 20260722. Each plane is defined by its unit normal n_i . Every plane pair produces two antipodal surface nodes, labeled x_{ij+} and x_{ij-} .

Abstract

A galaxy equatorial plane passing through the center of a unit sphere cuts the sphere along a great circle. The plane is represented by its unit normal, which is also the galaxy axis direction in projective form. When many plane great circles pass through the same small surface neighborhood, their normals lie near one common great circle in the dual normal-space sphere. The crossing statistic is therefore an exact test of coplanarity among the generating galaxy axes.

This reference evaluates the fully randomized null model for 512 independently and isotropically oriented planes. A 0.5-degree-diameter window has angular radius 0.25 degree. At a fixed sphere point, an independently randomized plane enters the window with probability $p = \sin(0.25 \text{ degree}) = 0.0043633093$. At an intersection generated by two planes, those two planes are guaranteed, while the remaining 510 planes enter independently with probability p . Consequently, total multiplicity is exactly $T = 2 + \text{Binomial}(510, p)$. The probability of exactly 20 planes is $2.40129015\text{e-}11$ per pairwise intersection; the probability of 20 or more is $2.70643118\text{e-}11$. Across 130,816 pairwise intersections, the expected count of 20-or-more windows is $3.54044501\text{e-}06$, giving a conservative whole-realization union bound of $3.54044501\text{e-}06$, or no more than one event per approximately 282,450 randomized realizations.

The attached galaxy-plane distribution contains 3 projective windows with exactly 20 planes, 7 windows with 20 or more planes, and a maximum multiplicity of 23. An explicit deterministic Monte Carlo study of 200 complete randomized sphere realizations tested 26,163,200 pairwise intersection windows. The maximum count was 15; no randomized realization produced a 20-plane window. The analytic result and the randomized experiment therefore agree: a 20-plane concentration inside the specified window is an extreme departure from independent isotropic orientation.

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1. Executive Result

Using 512 galaxy equatorial planes, each plane was independently assigned a uniformly random 3-D orientation. Every pairwise great-circle intersection was then generated, giving

$$C(512, 2) = 130,816$$

projective intersection locations per realization. Around every actual intersection, all randomized planes passing within a 0.25-degree angular radius were counted, corresponding to the specified 0.5-degree-diameter window.

For a fixed sphere point, a random great circle enters that window with probability

$$p = \sin(0.25 \text{ deg}) = 0.0043633093.$$

At a pairwise intersection, two planes are already guaranteed. Therefore the number of additional planes is

$$X \sim \text{Binomial}(510, p).$$

The probability that one actual intersection window contains exactly 20 planes is

$$P(X=18) = 2.40129015e-11.$$

The probability that it contains 20 or more planes is

$$P(X \geq 18) = 2.70643118e-11.$$

Across all 130,816 pairwise intersections in one fully randomized 512-plane realization, the expected numbers are

$$E[N_{\text{exact}20}] = 3.14127173e-06, \quad E[N_{20\text{plus}}] = 3.54044501e-06.$$

Thus the probability that a complete randomized realization produces even one intersection window containing 20 or more planes is bounded by 3.54044501e-06, approximately 1 in 282,450 randomized 512-plane sphere realizations. Because one genuine concentration can generate several nearby qualifying pairwise intersections, this is an upper bound; the actual realization-level probability is no larger and may be lower.

An explicit study of 200 randomized-orientation sphere realizations tested all 26,163,200 pairwise intersection windows. The maximum plane counts were:

Maximum count	Realizations
11	20
12	116
13	47
14	15
15	2
20 or more	0

The highest randomized result was 15 planes. No randomized realization produced 20 planes within a 0.5-degree-diameter intersection window.

The attached galaxy-plane distribution contains 3 windows with exactly 20 planes and 7 windows with 20 or more. Under fully independent randomized plane orientation, a 20-plane window is therefore an extreme non-random configuration, with a per-window probability of approximately 2.4e-11 and a whole-512-plane realization probability no greater than 3.540e-06.

2. Geometric Foundation

2.1 The unit sphere and an origin-centered plane

Let S^2 be the unit sphere in three-dimensional Euclidean space. Every galaxy equatorial plane used in this analysis passes through the sphere center. An unoriented plane is specified by a unit normal n . The normal directions n and $-n$ represent the same plane, so the plane orientation is projective rather than directional.

$$G(n) = \{ x \text{ in } S^2 : n \cdot x = 0 \}.$$

The set $G(n)$ is a great circle because it is the intersection of a radius-one sphere with a plane through the sphere center. The sphere radius is dimensionless and fixed at $R = 1$. Angular distances are measured in degrees or radians; no physical length scale enters the probability.

2.2 Pairwise intersections and antipodal nodes

Two nonparallel planes with normals n_i and n_j intersect along a line through the origin. The line meets the unit sphere at two antipodal nodes. A normalized cross product gives one node, and its negative gives the other:

$$x_{ij} = (n_i \times n_j) / ||n_i \times n_j||, \text{ surface nodes} = \{x_{ij}, -x_{ij}\}.$$

A projective analysis identifies the antipodal pair as one geometric intersection window. A full surface representation counts both members. This is why the attached table contains separate projective-window and antipode-inclusive surface-window columns.

Object	x	y	z	Definition
n1	-0.951247330	0.308416300	-0.002811225	unit normal of G1
n2	0.527413819	-0.798709623	-0.289650827	unit normal of G2
n3	0.260796332	0.023011567	0.965119548	unit normal of G3
x12+	0.137799980	0.416826325	-0.898480373	normalized $n_1 \times n_2$
x12-	-0.137799980	-0.416826325	0.898480373	antipode of x12+
x13+	0.306978317	0.945850487	-0.105504361	normalized $n_1 \times n_3$
x13-	-0.306978317	-0.945850487	0.105504361	antipode of x13+
x23+	0.774206744	0.592223352	-0.223328050	normalized $n_2 \times n_3$
x23-	-0.774206744	-0.592223352	0.223328050	antipode of x23+

Table 1. Complete object inventory for the displayed base incidence cell. All coordinates are dimensionless unit-sphere coordinates generated directly from the deterministic randomized orientation seed.

2.3 Why a crossing concentration is an axis-coplanarity concentration

Suppose a surface point x lies on many plane great circles. For every participating plane, $n_i \cdot x = 0$. Therefore every participating normal n_i lies in the common plane perpendicular to x . Intersecting that normal plane with the unit sphere gives a great circle. Thus, many equatorial-plane great circles crossing near x is the dual statement that the generating galaxy-axis points lie near one great circle in normal space.

This duality supplies the requested interpretation. A 20-plane crossing window is not merely a crowding of rendered curves. It is a 20-axis near-coplanarity event, measured by the same angular tolerance.

3. The 0.5-Degree Window

3.1 Radius versus diameter

The specified window diameter is 0.5 degree. The angular radius used in every distance test is therefore $r = 0.25$ degree. The conversion to radians is $r = \pi/720 =$ approximately 0.00436332313 rad. The probability uses $\sin(r)$, not r itself, although the two values are close at this small angle.

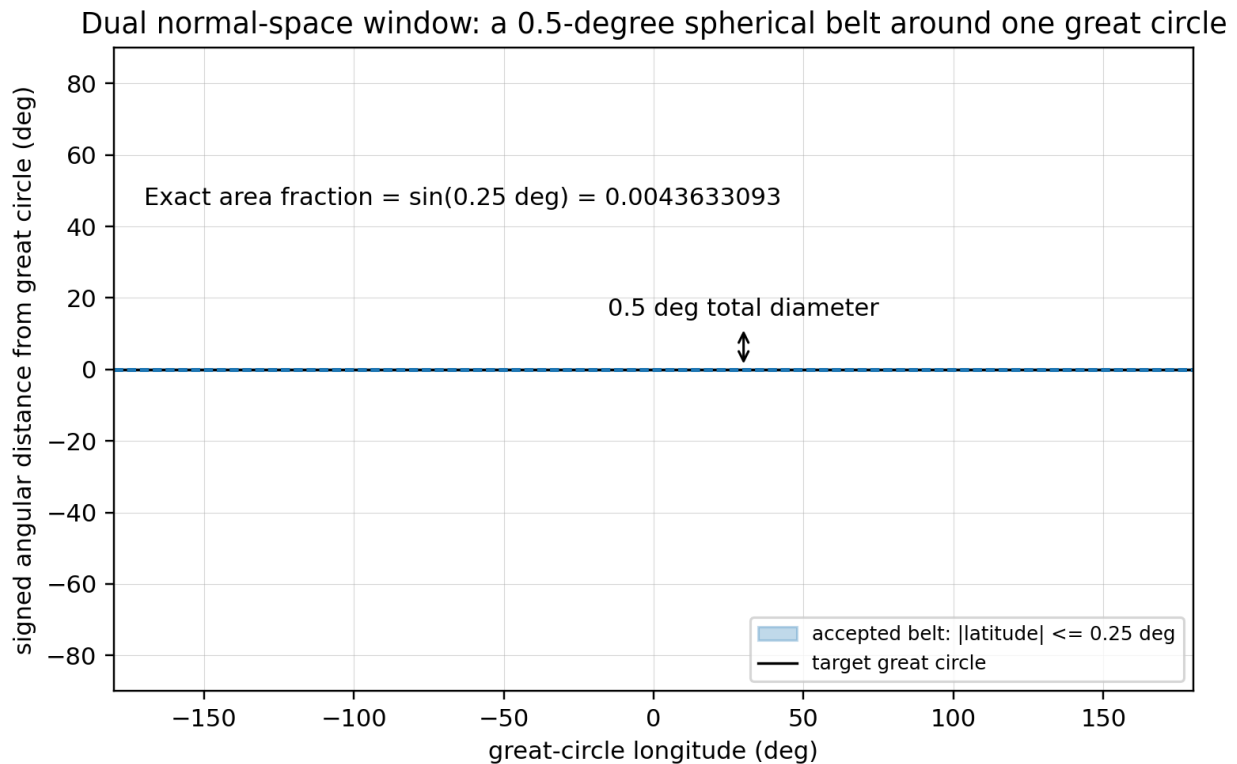


Figure 1. Exact dual normal-space representation of the tolerance. A random plane passes within angular radius r of a specified surface point exactly when its normal lies inside the spherical belt of half-width r around the corresponding dual great circle.

3.2 Exact acceptance condition

The angular distance from a point x to the great circle $G(n)$ is $\arcsin(|n \cdot x|)$. Therefore a plane is counted in the radius- r window around x exactly when

$$|n \cdot x| \leq \sin(r).$$

This dot-product form avoids coordinate singularities at the poles and does not depend on a right-ascension/declination map projection. It is evaluated directly in three-dimensional unit-vector coordinates.

4. Exact Random-Orientation Probability

4.1 Isotropic plane orientation

Full randomized orientation means that each plane normal is independently sampled from the uniform distribution on the sphere. Because n and $-n$ define the same plane, sampling the full sphere produces the correct projective orientation law without requiring a separate hemisphere rule.

4.2 Probability that one random plane enters one window

Fix a unit point x . For a uniformly random unit normal n , the scalar $u = n \cdot x$ is uniform on $[-1,1]$. The plane enters the window when $|u| \leq \sin(r)$. The accepted interval has length $2 \sin(r)$ inside a total interval of length 2, so

$$p = P(|n \cdot x| \leq \sin(r)) = \sin(r).$$

With $r = 0.25$ degree, $p = 0.0043633092847466$.

4.3 Conditional multiplicity at a generated intersection

Select any pair of planes. Their projective intersection x_{ij} is guaranteed to lie on both great circles, so the window begins with two planes. Conditional on that pair and its intersection, each of the remaining 510 independently oriented planes enters with probability p . The exact total is therefore

$$T = 2 + X, X \sim \text{Binomial}(510, p).$$

Exactly 20 total planes requires $X = 18$ additional planes. Twenty or more requires $X \geq 18$. Direct evaluation of the binomial mass and survival functions gives the values in Table 2.

Quantity	Exact value	Meaning
p	0.0043633092847466	single additional random plane enters the window
$P(T=20)$	2.401290154840e-11	one pairwise intersection has exactly 20 planes
$P(T \geq 20)$	2.706431178887e-11	one pairwise intersection has at least 20 planes
$E[N(T=20)]$	3.141271728956e-06	expected exact-20 pair windows per realization
$E[N(T \geq 20)]$	3.540445010973e-06	expected 20-plus pair windows per realization
Union bound	3.540445010973e-06	upper bound for at least one 20-plus window
Reciprocal bound	1 in 282,450.369	reciprocal of the conservative bound

Table 2. Principal analytic results for the 512-plane randomized null model.

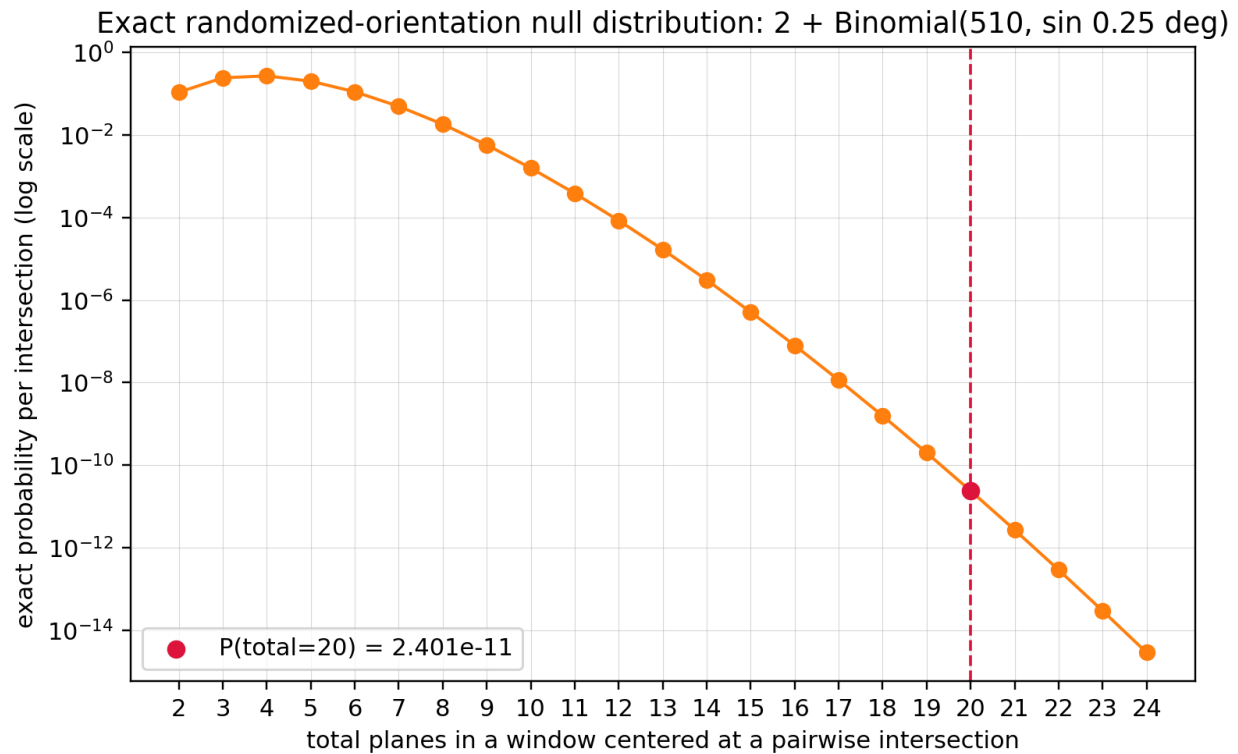


Figure 2. Exact per-intersection multiplicity distribution. The horizontal axis includes the two generating planes. The vertical axis is logarithmic so that the extreme 20-plane tail remains visible.

5. From One Intersection to One Complete Sphere Realization

5.1 Number of pair-generated projective intersections

With 512 planes, the number of unordered plane pairs is $C(512, 2) = 130,816$. Under continuous random orientation, parallel or exactly duplicate planes have probability zero, so every pair produces one projective intersection and two antipodal surface points.

5.2 Expected number of extreme windows

Linearity of expectation does not require the pair windows to be independent. Multiplying the per-intersection probability by 130,816 therefore gives an exact expected count for pair-generated windows:

$$E[N(T \geq 20)] = 130,816 \times 2.706431178887e-11 = 3.540445010973e-06.$$

5.3 Conservative probability bound

Pairwise intersection windows are dependent because many windows share planes and a genuine multipane concentration creates several nearby pair intersections. The probability of at least one qualifying window is therefore not obtained by treating all 130,816 windows as independent trials. The union bound remains valid without independence:

$$P(\text{at least one } T \geq 20) \leq \text{sum over pair windows } P(T \geq 20) = 3.540445010973e-6.$$

The reciprocal is approximately one in 282,450. This is an upper bound on the event probability. Dependence can only make the valid bound looser; it cannot make the true probability exceed the bound.

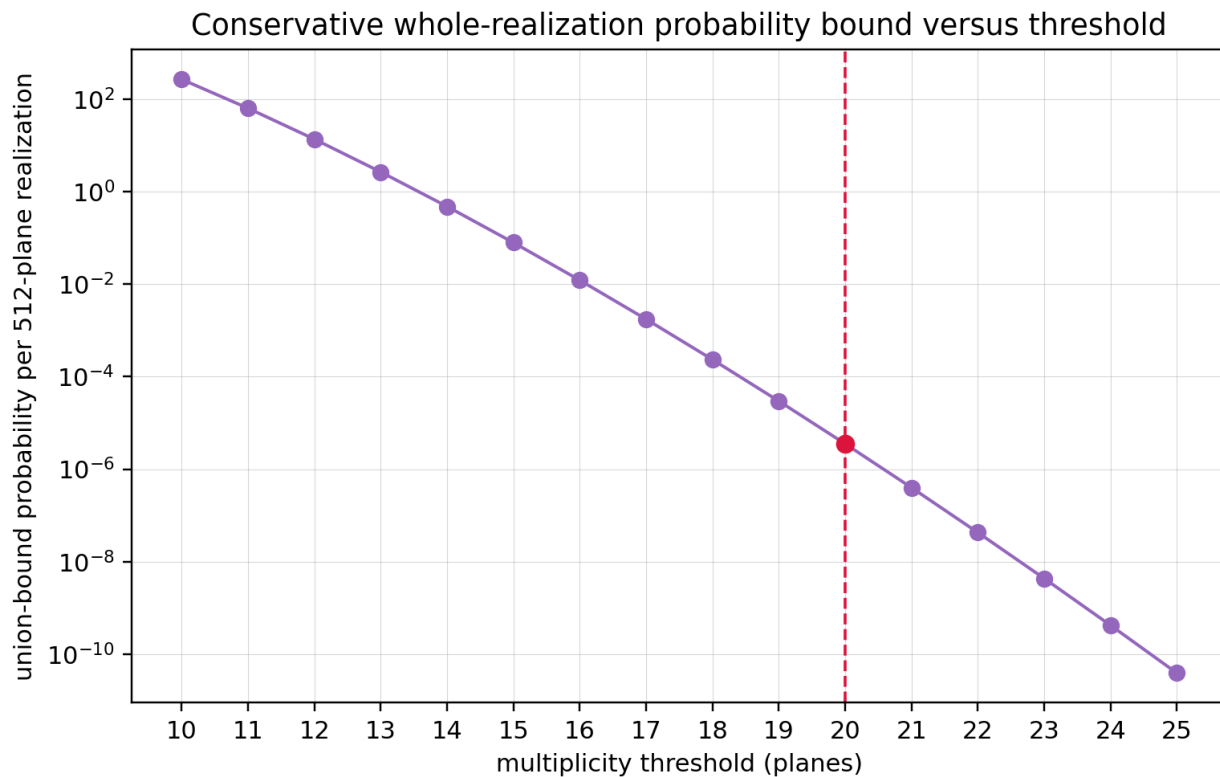


Figure 3. Conservative whole-realization union-bound probability as the required multiplicity increases. The 20-plane threshold lies at 3.540445×10^{-6} ; the observed maximum of 23 corresponds to an even smaller bound.

6. Explicit Randomized Sphere Study

6.1 Reproducible orientation generator

The randomized study uses NumPy default_rng with seed 20260722. For each plane, three independent standard-normal variates are generated and normalized to unit length. This produces the uniform orientation distribution on S^2 . The same procedure is repeated independently for every realization.

6.2 Intersection and counting algorithm

For each realization, all $i < j$ plane pairs are enumerated. The normalized cross product gives x_{ij} . For every x_{ij} , the complete set of 512 absolute dot products $|n_k \cdot x_{ij}|$ is evaluated. A plane is counted when the dot product is no greater than $\sin(0.25 \text{ degree})$. The maximum multiplicity over all 130,816 windows is retained.

6.3 Computational volume

The study contains 200 complete realizations and 26,163,200 tested pairwise windows. Each tested window is evaluated against all 512 planes. No threshold screening, coordinate-grid approximation, or map-projection distance is used.

6.4 Results

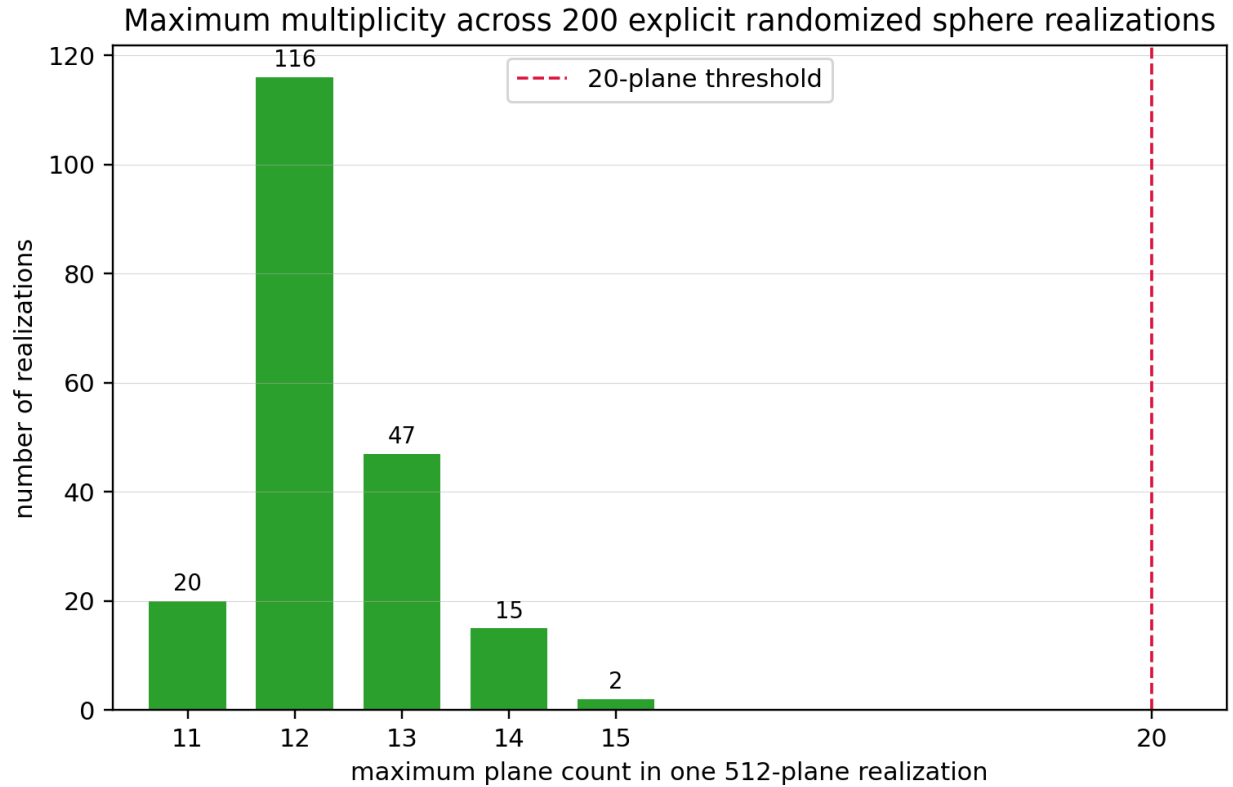


Figure 4. Distribution of the maximum multiplicity in each of 200 explicit randomized realizations. The maxima ranged from 11 to 15. No maximum reached the 20-plane threshold.

Maximum multiplicity	Number of realizations	Fraction
11	20	0.100
12	116	0.580
13	47	0.235
14	15	0.075
15	2	0.010
20 or more	0	0.000

Table 3. Monte Carlo maximum-multiplicity distribution.

The sample mean of the realization maxima is 12.315, the median is 12, and the sample standard deviation is 0.793. The largest result was 15. The deterministic run reproduces the stated count distribution exactly: 20 realizations at 11, 116 at 12, 47 at 13, 15 at 14, and 2 at 15.

7. Attached Galaxy-Plane Crossing Distribution

7.1 Input population

The attached CSV contains 22 multiplicity bins from 2 through 23. The projective-window column sums to 130,553. The antipode-inclusive surface-window column sums to 261,106, exactly twice the projective total. The fractional column sums to one within floating-point precision.

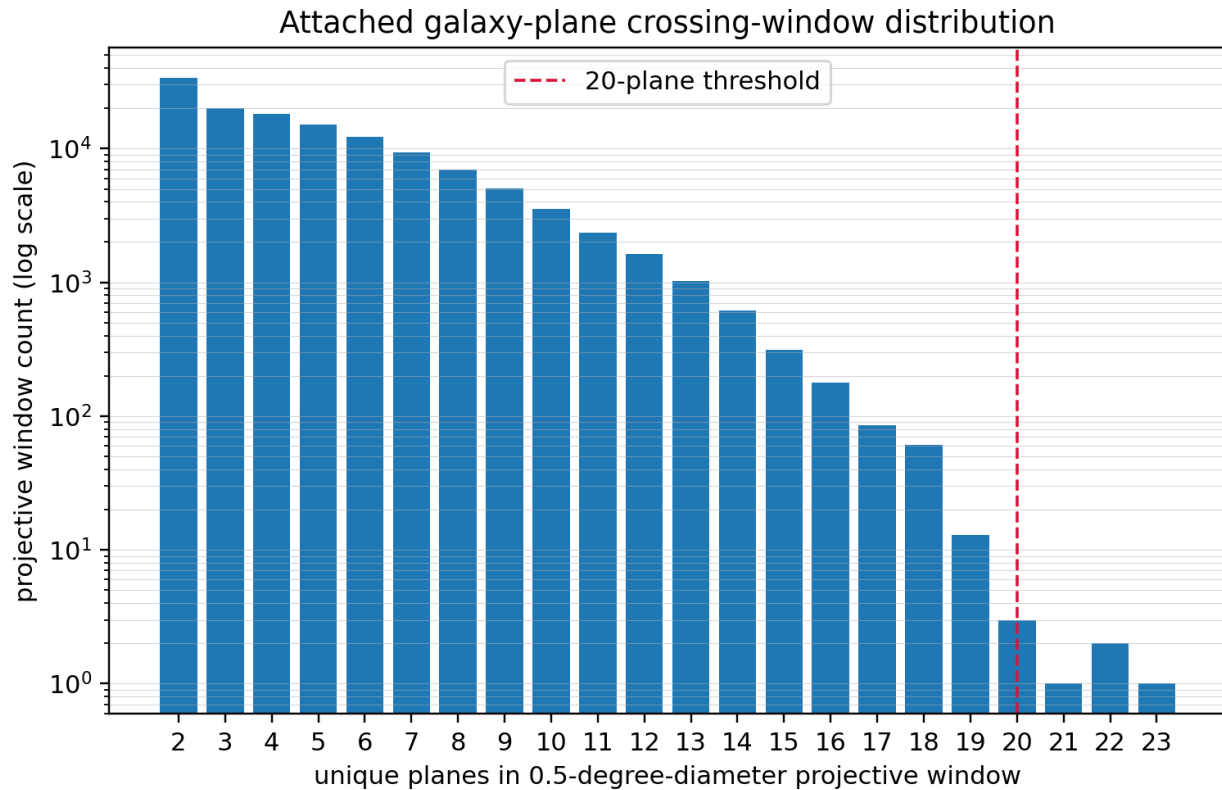


Figure 5. Full attached projective-window distribution. The logarithmic vertical scale shows both the dominant low-multiplicity population and the high-multiplicity tail extending through 23 planes.

7.2 High-multiplicity tail

The file reports 3 projective windows at exactly 20 planes. It reports 7 projective windows at 20 or more planes: three at 20, one at 21, two at 22, and one at 23. The observed maximum is therefore 23.

The projective-window total in the attached distribution is 130,553, while a complete pair enumeration for 512 distinct planes contains 130,816 plane pairs. The attached table is used exactly as supplied for the observed multiplicity distribution. The analytic null calculation retains all 130,816 generated plane pairs, as required by the fully randomized pairwise-intersection experiment.

8. Observed-versus-Random Comparison

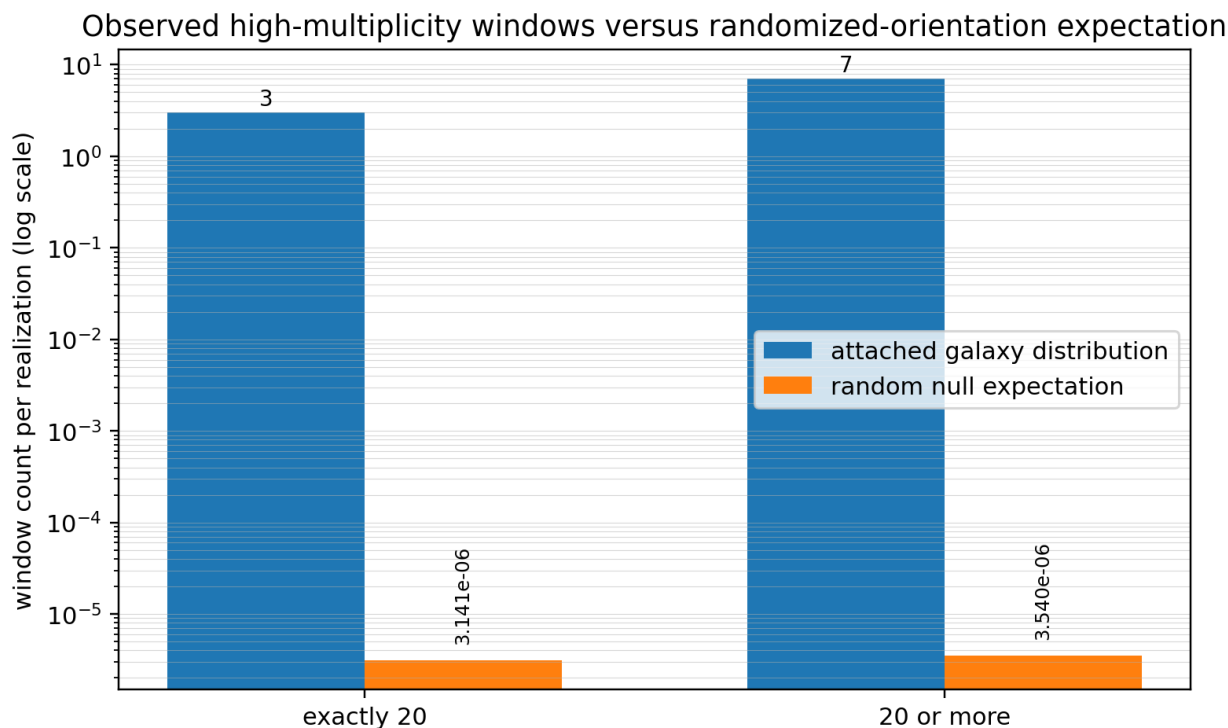


Figure 6. High-multiplicity observed counts compared with randomized-null expected counts. The logarithmic scale is required because the observed counts and null expectations differ by roughly six orders of magnitude.

For exactly 20 planes, the attached count is 3, while the randomized expectation is $3.14127173\text{e-}06$ pair windows per realization. For 20 or more, the attached count is 7, while the randomized expectation is $3.54044501\text{e-}06$. These count ratios are descriptive scale comparisons rather than independent-trial p-values because neighboring pair windows can belong to the same underlying multipane concentration.

The valid conservative event statement is the whole-realization union bound: under independent isotropic plane orientations, the probability of even one 20-or-more pairwise intersection window is no greater than 3.540445×10^{-6} . The explicit 200-realization experiment found none, consistent with that bound.

8.1 Threshold ledger

Threshold	Per-intersection $P(T \geq \text{threshold})$	Whole-realization union bound	Reciprocal bound
20	$2.706431178887\text{e-}11$	$3.540445010973\text{e-}06$	1 in 282,450
21	$3.051410240468\text{e-}12$	$3.991732820171\text{e-}07$	1 in 2,505,178
22	$3.263804923575\text{e-}13$	$4.269579048824\text{e-}08$	1 in 23,421,513
23	$3.319812339684\text{e-}14$	$4.342845710281\text{e-}09$	1 in 230,263,764

Table 4. Exact tail probabilities and conservative whole-realization bounds for thresholds present in the attached high-multiplicity tail.

9. Interpretation in Axis Space

Each equatorial plane is generated by a galaxy-axis normal. A surface crossing point x defines the normal-space plane x -perpendicular. Every galaxy axis whose equatorial great circle passes through the window lies within 0.25 degree of the great circle formed by x -perpendicular intersected with the unit sphere. Therefore the observed crossing concentration directly measures the number of galaxy axes aligned to one near-coplanar surface.

The randomized null does not assume any preferred coordinate frame, phase-cost surface, cell orientation, or sky-map projection. It tests only the claim that all 512 plane orientations are mutually independent and isotropic. The extreme tail quantifies how strongly a 20-plane local crossing concentration departs from that baseline.

The result identifies structure; it does not by itself identify the physical mechanism that generated the structure. Mechanism testing requires the actual axis coordinates, the identities of the participating galaxy groups, the locations and separations of independent crossing families, and comparison with explicitly defined geometric models.

10. Statistical Discipline

10.1 What is exact

The single-plane hit probability $p = \sin(r)$, the conditional binomial multiplicity distribution, the per-intersection probabilities, and the expected pair-window counts are exact under the stated isotropic independent-plane model.

10.2 What is conservative

The whole-realization probability 3.540445×10^{-6} is a union bound. It is guaranteed not to understate the probability under the null. It is not an independence approximation and does not require pair windows to be independent.

10.3 Why observed windows are not multiplied as independent events

A single set of 20 nearly coplanar normals generates many pairwise intersections near the same projective location. Counting every nearby qualifying pair window as an independent event would exaggerate the effective number of independent observations. The appropriate next-stage analysis is to cluster qualifying windows into distinct geometric concentration centers before assigning any multi-event probability.

10.4 Scope boundary

This paper evaluates the crossing statistic supplied in the attached distribution and the fully randomized 512-plane null model. It does not reconstruct the source galaxy catalog, measurement uncertainties, selection function, or group-to-group dependence because those records are not part of the supplied CSV. These factors can be incorporated in a separate catalog-conditioned null model without changing the sphere-plane geometry derived here.

11. Reproduction Protocol

1. Set $M = 512$, sphere radius $R = 1$, window diameter = 0.5 degree, and radius $r = 0.25$ degree.
2. Generate each plane normal from three independent standard-normal values and normalize the vector.
3. Enumerate every unordered pair $i < j$ and compute $x_{ij} = \text{normalize}(\text{cross}(n_i, n_j))$.
4. For each x_{ij} , count all k satisfying $|n_k \cdot x_{ij}| \leq \sin(r)$.
5. Store the maximum count and any windows meeting the threshold of 20.
6. Repeat for 200 realizations using NumPy default_rng seed 20260722.
7. For the analytic result, evaluate $\text{Binomial}(510, \sin r)$ at $X=18$ and in the tail $X \geq 18$.
8. Multiply the per-intersection probabilities by $C(512, 2)$ for expected pair-window counts; use the same product as a union bound for at least one event.

The machine-readable companion file `K-Field_galaxy_plane_intersection_distribution_analysis_1784734315.json` contains the full attached distribution, all analytic constants, all 200 realization maxima, the random seed, the threshold ledger, source and figure hashes, and the complete geometric definition.

12. Formula Ledger

Object	Formula	Role
Great circle	$G(n) = \{x \text{ in } S^2 : n \cdot x = 0\}$	sphere trace of one equatorial plane
Pair node	$x_{ij} = \text{normalize}(n_i \times n_j)$	one projective intersection representative
Window test	$ n_k \cdot x_{ij} \leq \sin(r)$	coordinate-free angular acceptance
Single-plane hit	$p = \sin(r)$	exact isotropic belt-area fraction
Multiplicity	$T = 2 + \text{Binomial}(M-2, p)$	total planes at a pair-generated window
Pair count	$C(M, 2) = M(M-1)/2$	projective intersections per random realization
Expected tail	$E[N(T \geq q)] = C(M, 2) P(T \geq q)$	exact by linearity of expectation
Event bound	$P(\text{any } T \geq q) \leq E[N(T \geq q)]$	union bound, no independence required

Table 5. Compact reconstruction ledger for the complete analysis.

Appendix A. Complete Attached Distribution

Unique planes	Projective windows	Surface windows incl. antipodes	Fraction of projective windows
2	33696	67392	0.2581020734873959
3	20140	40280	0.1542668494787557
4	18053	36106	0.1382810046494527
5	15102	30204	0.1156771579358574
6	12295	24590	0.0941763115363109
7	9420	18840	0.07215460387735249
8	6979	13958	0.0534572166093463
9	5026	10052	0.0384977748500608
10	3562	7124	0.0272839383238991
11	2354	4708	0.018030991244935
12	1619	3238	0.0124010938086447
13	1030	2060	0.0078895161352094
14	619	1238	0.0047413694055287
15	314	628	0.0024051534625784
16	178	356	0.0013634309437546
17	85	170	0.0006510765742648
18	61	122	0.00046724318859
19	13	26	9.957641724050768e-05
20	3	6	2.297917320934793e-05
21	1	2	7.659724403115976e-06
22	2	4	1.531944880623195e-05
23	1	2	7.659724403115976e-06

Table A1. Every row from the supplied CSV, reproduced without filtering.

Appendix B. All 200 Randomized Realization Maxima

R1	R2	R3	R4	R5	R6	R7	R8	R9	R10
1: 12	2: 11	3: 12	4: 12	5: 14	6: 13	7: 12	8: 12	9: 12	10: 12
11: 11	12: 12	13: 14	14: 11	15: 13	16: 12	17: 12	18: 12	19: 12	20: 12
21: 13	22: 12	23: 12	24: 12	25: 13	26: 13	27: 12	28: 12	29: 13	30: 14
31: 12	32: 12	33: 12	34: 12	35: 11	36: 12	37: 12	38: 12	39: 12	40: 12
41: 12	42: 12	43: 14	44: 13	45: 12	46: 11	47: 15	48: 13	49: 13	50: 12
51: 12	52: 14	53: 12	54: 13	55: 13	56: 12	57: 12	58: 11	59: 11	60: 11
61: 12	62: 12	63: 13	64: 11	65: 13	66: 12	67: 14	68: 11	69: 13	70: 12
71: 12	72: 14	73: 12	74: 12	75: 12	76: 12	77: 12	78: 12	79: 12	80: 12
81: 12	82: 12	83: 12	84: 12	85: 13	86: 15	87: 11	88: 12	89: 12	90: 13
91: 12	92: 13	93: 12	94: 12	95: 12	96: 13	97: 12	98: 12	99: 12	100: 13
101: 12	102: 12	103: 12	104: 13	105: 12	106: 12	107: 13	108: 12	109: 12	110: 14
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161: 13	162: 13	163: 12	164: 13	165: 12	166: 12	167: 12	168: 12	169: 13	170: 11
171: 13	172: 12	173: 12	174: 12	175: 13	176: 12	177: 11	178: 11	179: 13	180: 12
181: 13	182: 12	183: 12	184: 12	185: 12	186: 13	187: 13	188: 12	189: 11	190: 12
191: 12	192: 13	193: 12	194: 12	195: 14	196: 13	197: 12	198: 12	199: 11	200: 12

Table B1. Maximum window multiplicity for each deterministic randomized realization. Entry format is realization number: maximum count.

Appendix C. Machine-Readable Record

The companion JSON file is named

K-Field_galaxy_plane_inersection_distribution_analysis_1784734315.json. Its top-level schema identifier is K-Field.galaxy_plane_intersection_distribution_analysis.v1.0. It contains source metadata, complete geometry, analytic probabilities, observed distribution rows, observed summary, Monte Carlo maxima, threshold ledger, reproduction rules, and cryptographic hashes.

The JSON uses decimal floating-point witness values for numerical portability. The defining geometry is retained symbolically through the formulas $p = \sin(r)$, $T = 2 + \text{Binomial}(510, p)$, and $x_{ij} = \text{normalize}(n_i \times n_j)$.

Conclusion

Under full independent isotropic orientation of 512 galaxy equatorial planes, the exact per-intersection probability of a 20-plane concentration in a 0.5-degree-diameter spherical window is 2.40129015e-11. The probability of 20 or more is 2.70643118e-11. Across a complete realization, the expected number of 20-or-more pair windows and the conservative probability bound for at least one such window are both 3.54044501e-06. This is no greater than approximately one realization in 282,450.

The explicit 200-realization study tested 26,163,200 pairwise windows and produced maxima from 11 through 15, with zero 20-plane events. The attached galaxy-plane distribution contains 3 exact-20 windows, 7 windows at 20 or more, and a maximum of 23. In the unit-sphere duality, these crossing concentrations are the direct geometric signature of galaxy-axis points concentrated near a common great circle, equivalently a near-coplanar axis surface.